do280-4.18: Days remaining **89** | Lab hours used **4/80**

DO280-4.22: Days remaining **89** | Lab hours used **0/80**

Red Hat OpenShift Administration II: Configuring a Production Cluster

Course | Lab Environment

Lab: Deploy Packaged Applications

Deploy a Helm chart.

Deploy an application from an OpenShift template.

Deploy an application with Kustomize.

Outcomes

- Deploy applications from Helm charts.
- Deploy an application from a template.
- Customize an application from a Kustomize layer.
- Access the applications over HTTPS.



NOTE

The allocated time for this activity is 45 minutes. If you need additional time to

complete the task, then you must revisit the course content, or practice more.

As the `student` user on the `workstation` machine, use the `lab` command to prepare your environment for this exercise, and to ensure that all required resources are available:

```
[student@workstation ~]$ lab start compreview-package
```

The `lab start` command copies all required files for this exercise to the `~/DO0019L/labs/compreview-package` directory.

Specifications

The company wants to deploy the Etherpad collaborative editor to support its internal documentation and real-time collaboration needs. You must deploy the application across multiple environments, such as development and production.

As a developer in this organization, your responsibility is to deploy the Etherpad application by using a Helm chart in both environments. By using Kustomize, you update the application in the production environment with specific configurations.

The cluster API URL is `https://api.lab.example.com:6443`. The password for the `developer` user is `developer`, and the password for the `admin` user is `redhatocp`.

> **NOTE**
>
> You can create the necessary files in the `~/DO0019L/labs/compreview-package` directory.

- Add the `classroom` repository from the `http://helm.lab.example.com/charts` URL.
- The `etherpad-dev` project hosts the development environment of the Etherpad application.
- The `etherpad-prod` project hosts the production environment of the Etherpad application.
- Deploy the `0.0.7` version of the Etherpad chart in the development environment.
  - The Helm release name is `etherpad-dev`.
  - The application uses the `registry.lab.example.com:8443/etherpad` image, with the `1.8.18` image tag, and `etherpad` as the image name.
  - The deployment has one replica.
  - When the application is deployed, you can access it at the `https://etherpad-dev.apps.lab.example.com` URL.
- Deploy the `0.0.7` version of the Etherpad chart in the production environment.
  - The Helm release name is `etherpad-prod`.

  - The application uses the `registry.lab.example.com:8443/etherpad` image, with the `1.8.18` image tag, and `etherpad` as the image name.
  - The deployment has three replicas.
  - When the application is deployed, you can access it at the `https://etherpad-prod.apps.lab.example.com` URL.

- Deploy a PostgreSQL database in the production environment by using the `postgresql-persistent` template.
  - The database name is `etherpad`.
  - The database user is `etherpad`.
  - The database password is `redhatocp123`.
  - The database volume capacity is `1Gi`.
- The `postgresql-persistent` template is in the `openshift` namespace.
- The `postgresql-persistent` template creates a `postgresql` secret with the database credentials that you use to configure the Etherpad application.
- The `lab` command creates the Kustomize base overlay for the Etherpad application in the `~/DO0019L/labs/compreview-package/etherpad` directory.
- Add a new Kustomize `production` overlay that uses the `~/DO0019L/labs/compreview-package/kustomization.yaml` file, and the `PodDisruptionBudget` resource in the `~/DO0019L/labs/compreview-package/pdb.yaml` file.
- The `production` overlay has the following configurations:

Key	Value
`app.kubernetes.io/environment`	`production`
`managed-by`	`kustomize-gitops`
`spec.replicas`	`6`
`requests.cpu`	`200m`
`requests.memory`	`256Mi`
`limits.cpu`	`500m`
`limits.memory`	`512Mi`

- To connect the Etherpad application to the PostgreSQL database, add the following environment variables to the Etherpad deployment by using the Kustomize overlay:

Environment Variable	Value
DB_TYPE	postgres
DB_HOST	postgresql
DB_PORT	5432
DB_NAME	etherpad
DB_USER	Use the database-user key from the postgresql secret.
DB_PASS	Use the database-password key from the postgresql secret.

- Verify that the production environment configuration complies with the following settings:
  - The Etherpad application runs six pods.
  - The PostgreSQL database runs one pod.
  - All resources carry the required labels and annotations.
  - The application pods set the correct values for the resource requests and limits.
  - The application pods have the correct environment variables to connect to the PostgreSQL database.

> **NOTE**
>
> The lab command copies the solution files for the exercise to the ~/DO0019L/solutions/compreview-package directory.

1. Log in to the cluster as the developer user with developer as the password.
2. Add the classroom Helm repository from the http://helm.lab.example.com/charts URL.
3. Create the etherpad-dev and etherpad-prod projects.
4. Deploy Etherpad to the development environment.
   4.1. Create a file named ~/DO0019L/labs/compreview-package/dev-values.yaml with the following content:

```
image:
  repository: registry.lab.example.com:8443/etherpad
  name: etherpad
  tag: 1.8.18
route:
  host: etherpad-dev.apps.lab.example.com
```

4.2. Install the `0.0.7` version of the `classroom/etherpad` chart in the `etherpad-dev` project. Use the values file from the previous step.

5. Deploy Etherpad to the production environment.

5.1. Create a file named `~/DO0019L/labs/compreview-package/prod-values.yaml` with the following content:

```
image:
  repository: registry.lab.example.com:8443/etherpad
  name: etherpad
  tag: 1.8.18
route:
  host: etherpad-prod.apps.lab.example.com
replicaCount: 3
```

5.2. Install the `0.0.7` version of the `classroom/etherpad` chart to the `etherpad-prod` project. Use the values file from the previous step.

6. Deploy a PostgreSQL database in the production environment by using the `postgresql-persistent` template.

6.1. Use the `oc process` command to generate the manifests for the `postgresql-persistent` template, and use the `oc apply` command to create the resources in the cluster.

7. Prepare the resources for the Kustomize production overlay.

7.1. Copy the `~/DO0019L/labs/compreview-package/kustomization.yaml` file to the `production` overlay directory.

7.2. Copy the `~/DO0019L/labs/compreview-package/pdb.yaml` file to the `production` overlay directory.

7.3. Edit the `kustomization.yaml` file in the `production` overlay directory to match the production environment specifications and to connect the application to the PostgreSQL database.

8. Apply the production overlay, which replaces the Helm-managed release with the Kustomize-customized version.

9. Verify that the production deployment meets all requirements.

9.1. Check the pod count by running the following command:

```
[student@workstation ~]$ oc get pod -n etherpad-prod \
| grep etherpad-prod | wc -l
6
```

9.2. Verify that the PostgreSQL database is running.

9.3. Check that the labels and annotations on the resources in the `etherpad-prod` project match the ones that Kustomization applies.

9.4. Verify that the `PodDisruptionBudget` resource exists in the `etherpad-prod` project.

9.5. Verify that the application pods have the correct environment variables to connect to the PostgreSQL database.

10. Verify that the Etherpad application is accessible in both environments by using the following URLs:

- Development URL: `https://etherpad-dev.apps.lab.example.com`
- Production URL: `https://etherpad-prod.apps.lab.example.com`

Hide Solution

Evaluation

Evaluate the allocated time for this activity against the time that you took to complete it. If you used more time than was allocated, then you must revisit the course content, or practice more.

> **NOTE**
>
> The grading scripts provide remediation hints to help you identify what content you must revisit to complete the tasks.

As the `student` user on the `workstation` machine, use the `lab` command to grade your work. Correct any reported failures and rerun the command until successful.

```
[student@workstation ~]$ lab grade compreview-package
```

Finish

As the `student` user on the `workstation` machine, use the `lab` command to complete this exercise. This step is important to ensure that resources from previous exercises do not impact upcoming exercises.

```
[student@workstation ~]$ lab finish compreview-package
```

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